



MODULE 3 ASSIGNMENT RUBRIC

Mission Operations Report: Managing a Crop to Harvest

Applies to the Module 3 assignment only

Total: 100 points

CLOs: 4, 5, 6

Length: 350–450 words

Type: Analytic · Weighted criteria

CRITERIA	EXCELLENT 100–90%	PROFICIENT 89–75%	DEVELOPING 74–60%	BEGINNING BELOW 60%
Growth Cycle Overview 15% of total	Accurately names the crop and describes its growth cycle with specific detail -- the number of phases, approximate duration of each, and what the crop needed to transition successfully from one phase to the next. Shows clear understanding of how a full growth cycle works in a space environment.	Names the crop and describes the growth cycle adequately but with less precision. Phase durations or requirements may be approximate. Demonstrates general understanding without phase-specific detail.	Names the crop but growth cycle description is vague or incomplete. Phases are mentioned without describing their requirements or duration.	Crop is named without meaningful description of the growth cycle, or the description contains significant inaccuracies about how the crop grows.
Critical Phase Transition Analysis 35% of total	Identifies two specific phase transitions and for each describes: which variables were adjusted, the reasoning behind each adjustment, and what actually happened as a result. Analysis is specific enough to be actionable -- not just "I adjusted the temperature" but what the value was changed to, why that value was chosen, and what the crop's response was.	Identifies two phase transitions and describes variable adjustments for each, but reasoning or outcomes are described for only one of the two. One transition is analyzed well while the other receives surface treatment.	Identifies phase transitions but variable adjustments are described without reasoning or outcomes. Reads like a log of what happened rather than an analysis of why decisions were made.	Phase transitions are not clearly identified, or the analysis does not describe specific variable adjustments. Does not demonstrate understanding of how managing variables at phase transitions affects crop outcomes.
What Worked and What Did Not 25% of total	Identifies one specific decision that worked well and one that did not, and for each provides a clear explanation of what the outcome revealed about managing a food system under mission constraints. Moves beyond "it worked" or "it failed" to describe what was learned and why that matters in a real mission context.	Identifies a success and a failure with reasonable explanations, but the connection to mission constraints is present without being fully developed. Describes what happened more than what it revealed.	Identifies a success and a failure but explanations are vague. Does not clearly connect the outcomes to the broader context of managing a food system under mission constraints.	Does not clearly identify both a success and a failure, or claims everything worked perfectly without acknowledging tradeoffs or constraints inherent in any design.
Recommendations for Next Crew 15% of total	Provides two concrete, actionable recommendations specific enough for the next crew to apply directly. Names specific variables, phase transitions, or decision points. For example: "During pollination, hold temperature between 78–83°F and do not drop below MED light even if vegetative growth has slowed" is actionable. "Watch the temperature carefully" is not.	Provides two recommendations that are relevant and generally useful but lack the specificity needed to be directly actionable. The next crew would need to interpret or refine the advice before applying it.	Recommendations are vague, generic, or obvious. Does not demonstrate that the recommendations were derived from actual experience managing this specific crop.	No recommendations provided, or recommendations are unrelated to the crop or growth cycle managed in the simulation.
Professional Document Format and Clarity 10% of total	Report is organized by the four required sections, written in paragraph form, and stays within the 350–450 word range. Writing is clear and reads as a professional mission document -- direct, specific, and useful to a reader who needs accurate information.	Report follows the required structure with minor organizational issues. Writing is mostly clear and professional. Stays close to the word range.	Report structure is present but inconsistent. Some sections read as lists or informal notes. May be significantly over or under the word range.	Report does not follow the required section structure, is not written in paragraph form, or is so far outside the word range that it does not meet the basic requirements of the assignment.

SCORE KEY ● Excellent: 90–100 pts ● Proficient: 75–89 pts ● Developing: 60–74 pts ● Beginning: Below 60 pts

EQUITY & ACCESSIBILITY NOTE

This rubric evaluates operational thinking and decision analysis, not prior knowledge of plant science or agriculture. A learner who has never grown a plant before but who engaged carefully with the Lunar Greenhouse simulation and can analyze what their decisions produced will score as well as someone with farming experience. The professional document format requirement is kept simple and clearly structured to reduce barriers for learners less familiar with formal writing conventions.